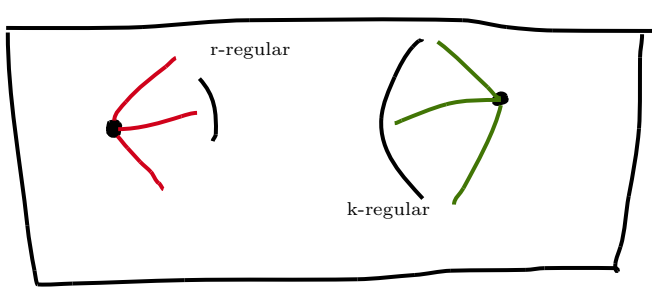

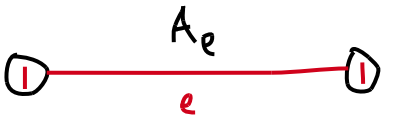

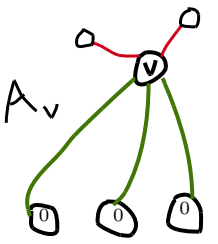

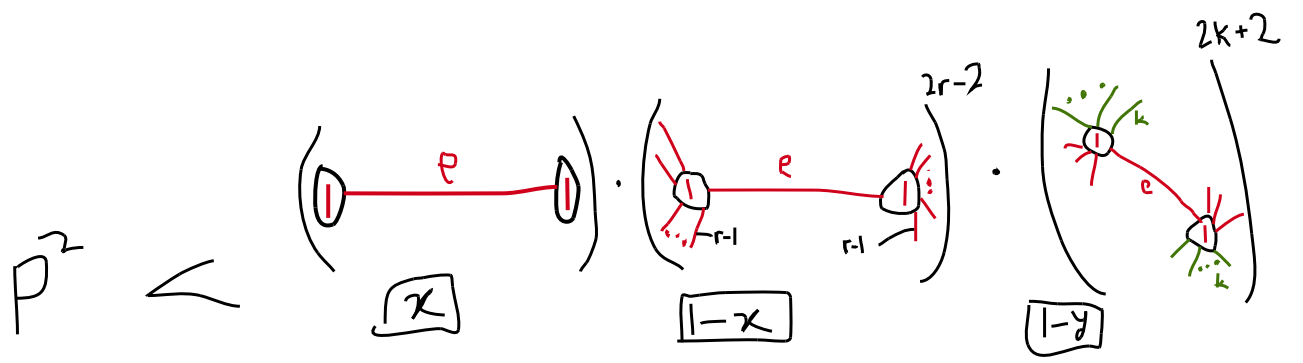

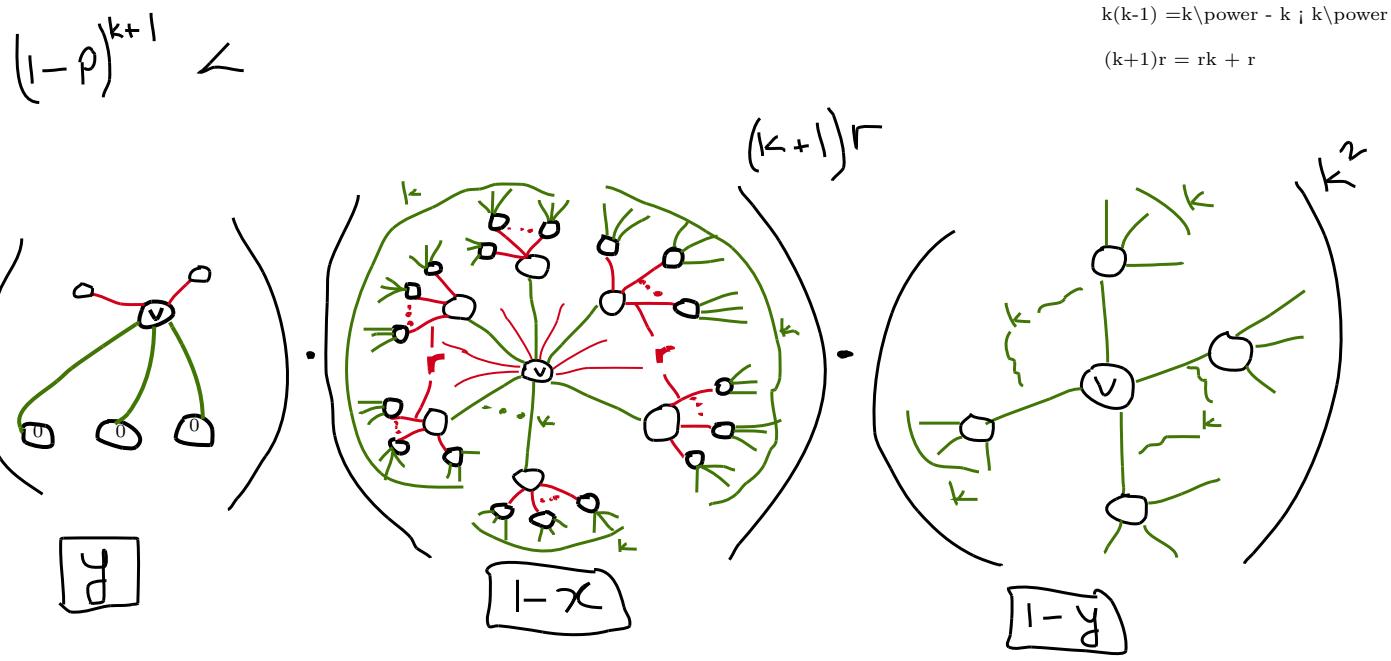

k(k-1) =k\power - k ¡ k\power

(k+1)r = rk + r

Thomassen's bounds not satisfying LLL for 4-regular graphs

Since we are constructing a set S that is neither red independent nor green dominating, we want the probabilities \math-container and \math-container to fail. Now let's compute the bounds for \math-container and \math-container. To do that assume \math-container and \math-container

We derive the values of \math-container by using \math-container where \math-container for \math-container and \math-container for \math-container.

$P; (171.16, 1419.86) node[anchor = northwest, inner sep = 0.75pt, align = left, font =]$P

p\power x (1-x)\power(1-y)\power
We know r=2 and k =2
p\power \frac (1-\frac)\power(1-\frac)\power
p\power \frac (\frac)\power(\frac)\power 0.0388
p 0.197
0.197\power = 0.651
0.651 0.038

(1-p)\power y (1-x)\power (1-y)\power\power
We know r=2 and k=2
(1-p)\power \frac (1-\frac)\power(1-\frac)\power
(1-p)\power \frac (\frac)\power(\frac)\power 0.007
(1-p) 0.007
(1-p) 0.19 p 0.8
(0.8)\power = 0.512
0.512 0.007

To find a second cycle we need to show the existence of \math-container, in order to do that we need to satisfy
LLL for both \math-container and \math-container. Since neither bound is staisfied it shows that no such \math-container exists to gurantee the existence of a second Hamiltonian cycle.
$-container$
Set \math-container exists \math-container \math-container such that \math-container